EXOPLANET

Project Report Submitted in fulfillment of the Requirements for the Award of the Internship of Winter Training Program on Space Science Technology

Astronomy and Astrophysics

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ABSTRACT

This project investigates the exoplanet TRAPPIST-1e, an Earth-sized world orbiting an ultra-cool M-dwarf star approximately 40 light-years away. The primary objective is to analyse photometric data from the Kepler Space Telescope's K2 mission to isolate the planet's transit signature and determine its fundamental physical properties. Using Python's lightkurve package, raw light curve data was processed to remove instrumental anomalies.

Calculations based on the transit parameters yielded a planetary density of 4.87 g/cm³, closely aligning with Earth's density of 5.51 g/cm³. This confirms the exoplanet's composition as a terrestrial, rocky body rather than a gas giant, effectively ruling out a thick, Primary hydrogen-rich atmosphere. Furthermore, with an estimated equilibrium temperature of about 250 K, TRAPPIST-1e resides within the host star's habitable zone. This suggests the potential to support liquid water on its surface, provided the presence of a heavier secondary atmosphere driven by greenhouse gases like carbon dioxide or nitrogen. Ultimately, this analysis validates the transit method's effectiveness in characterizing Earth-like exoplanets and reaffirms TRAPPIST-1e as a prime candidate for future Astro biological studies.

Introduction: TRAPPIST-1e

Host Star:

TRAPPIST-1e is an exoplanet orbiting a very cool red dwarf star TRAPPIST-1. The star is a M type dwarf, much cooler and less luminous than sun.

TRAPPIST-1 is approximately 40 light years from Earth.

Because its smaller and much cooler than our Sun, its habitable zone is much closer than compared to that of our Sun's.

Detection Method:

TRAPPIST-1e is discovered using **transit method.**

Here, we observe the periodic decrease in the star's brightness when the planet transits (passes in between the star and our view from earth).

Through this method we can determine Orbital Period, Transit depth, Planetary radius.

Orbital Period:

The orbital period of TRAPPIST-1e is approximately 6.1 Earth days.

TRAPPIST-1e is considered one of the most Earth-sized exoplanets discovered to date. It orbits within the habitable zone. Water might exist in liquid state. It is Earth-sized, density similar to Earth, MIGHT satisfy necessary conditions to sustain life. However, the intense stellar radiations from its host star might pose a challenge to sustain human life by threatening its atmosphere.

Physics Calculations

For the calculations I took the data of TRAPPIST-1e from NASA archive: https://exoplanetarchive.ipac.caltech.edu/ . Thus used the following formulas to find the asked parameters.

Transit Depth Formula:

$$R_p = R_\star \sqrt{\delta}$$

$$\delta \rightarrow \ Transit\ depth\,,$$

Error:

$$\left(\frac{\sigma_{R_p}}{R_p}\right)^2 = \left(\frac{\sigma_{R_*}}{R_*}\right)^2 + \frac{1}{4}\left(\frac{\sigma_{\delta_*}}{\delta_*}\right)^2$$

Equilibrium Temperature:

$$T_p = T_* \sqrt{\frac{R_*}{2a}}$$

Error:

$$\left(\frac{\sigma_T}{T}\right)^2 = \left(\frac{\sigma_{T_*}}{T_*}\right)^2 + \frac{1}{4}\left(\frac{\sigma_{R_*}}{R_*}\right)^2 + \frac{1}{4}\left(\frac{\sigma_a}{a}\right)^2$$

Density:

$$\rho = \frac{3M}{4\pi R^3}, \quad 1\,\rho_\oplus = 5.51 g/cm^2$$

Error:

$$\delta_a = \rho\left(\left(\frac{\sigma_M}{M}\right) + 3\left(\frac{\sigma_R}{R}\right)\right)$$

Quantity	Value	Uncertainity
Stellar Temperature (Teff (K))	2566	26
Stellar Radius $R_\star$ $(R_\odot)$	0.1192	0.0013
Stellar Mass $M_\star$ $(M_\odot)$	0.0898	0.0023
Semi-major axis a (au)	0.02925	0.0025
Planet Mass $M_\mathrm{p}(M_\oplus)$	0.692	0.022
Transit Depth (%)	0.5012	0.0078

Planet Radius

$$R_p = R_\star\sqrt{\delta}$$

$\delta \rightarrow$ Transit depth ,

Rp = 0.008438821 $R_\odot$

$1R_\odot = 109.1R_\oplus$

Rp = 0.920675396 $R_\oplus$

Error Calculation

$$\left(\frac{\sigma_{R_p}}{R_p}\right)^2 = \left(\frac{\sigma_{\mathrm{R}_*}}{\mathrm{R}_*}\right)^2 + \frac{1}{2}\left(\frac{\sigma_{\delta_*}}{\delta_*}\right)^2$$

0.014264223

Equilibrium Temperature

1R⊙ = 0.00465 au 0.000554

$$T_p = T_*\sqrt{\frac{R_*}{2a}}$$

Equilibrium Temp: Tp = 249.7718

Error

$$\left(\frac{\sigma_T}{T}\right)^2 = \left(\frac{\sigma_{T_*}}{T_*}\right)^2 + \frac{1}{4}\left(\frac{\sigma_{R_*}}{R_*}\right)^2 + \frac{1}{4}\left(\frac{\sigma_a}{\mathrm{a}}\right)^2$$

Error = 11.0541622

$$T_p = 249.722 \pm 11.0541622$$

Density

$$\rho = \frac{3\mathrm{M}}{4\pi\mathrm{R}^3}, \quad 1\,\rho_\oplus = 5.51 g/cm^2$$

pi = 3.141593

ρ = 4.872637778

$$\delta_a = \rho\left(\left(\frac{\sigma_M}{M}\right) + 3\left(\frac{\sigma_R}{R}\right)\right)$$

Error= 0.38138894

$$\rho = 4.87263778 \pm 0.3507527$$

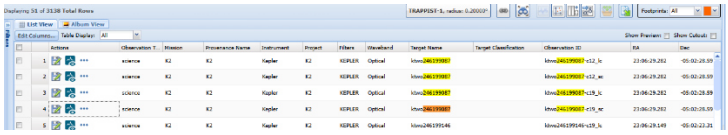
Parameter	Value	Uncertainity	Units
Radius	0.9206794	0.01233467	R
Temperature	249.772	11.041622	K
Density	4.87263778	0.3507527	g/cm^3

Earth Comparison Table

Parameter	Earth	TRAPPIST-1e
Temperature T_f	255	~250
Radius $(R_\oplus)$	1.00	~0.92
Orbital Period (days)	365	~6.1
Density	5.51	~4.87
Mass $M_\oplus$	1.00	0.69

Light Curve Analysis

For the analysis, firstly I took the light curve which I found in NASA Exoplanet Archives, https://mast.stsci.edu/portal/Mashup/Clients/Mast/Portal.html . Here I found the data of trappist-1 from the K2 Mission, file named: ktwo246199087. The data downloaded was in .fits format thus I had to convert it into csv and process it using python's lightkurve library, then clean the data, remove stellar noise, and finally plot flux versus time.



Important Terms:

Flux:

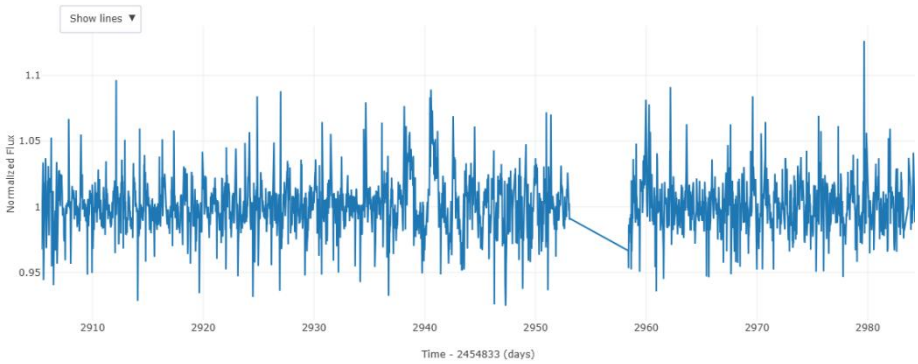
It is defined as the brightness of a star as measured by the telescope.

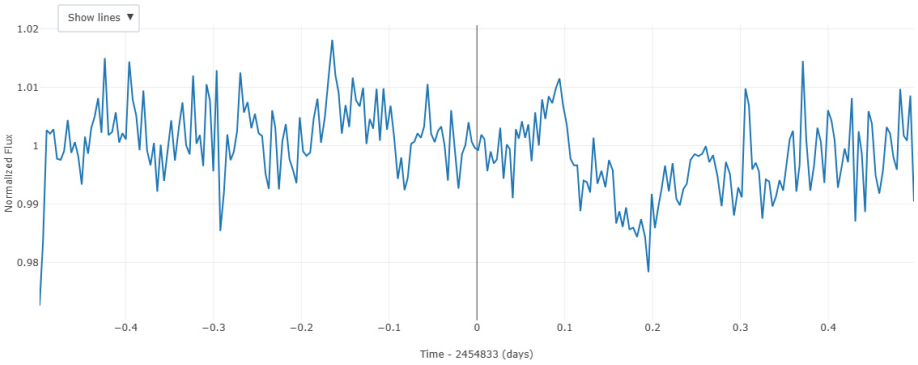
From my data set, the normalized flux is 1 (approx.).

Time Axis:

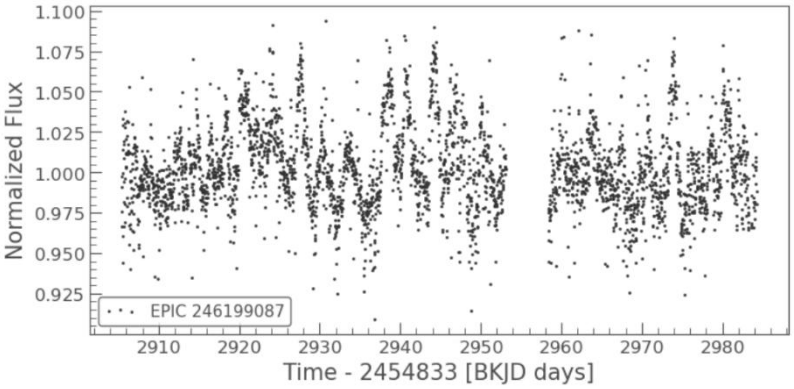
This represents the observation time of the telescope.

Measured in days.





[8]:
```
lc.scatter()
plt.show()
```



[9]:
```
x = np.percentile(lc.flux,1)
depth = 1-x
print("Transit depth:", depth)
```

```
Transit depth: 0.05612589657306666
```

[ ]:

Calculation Results

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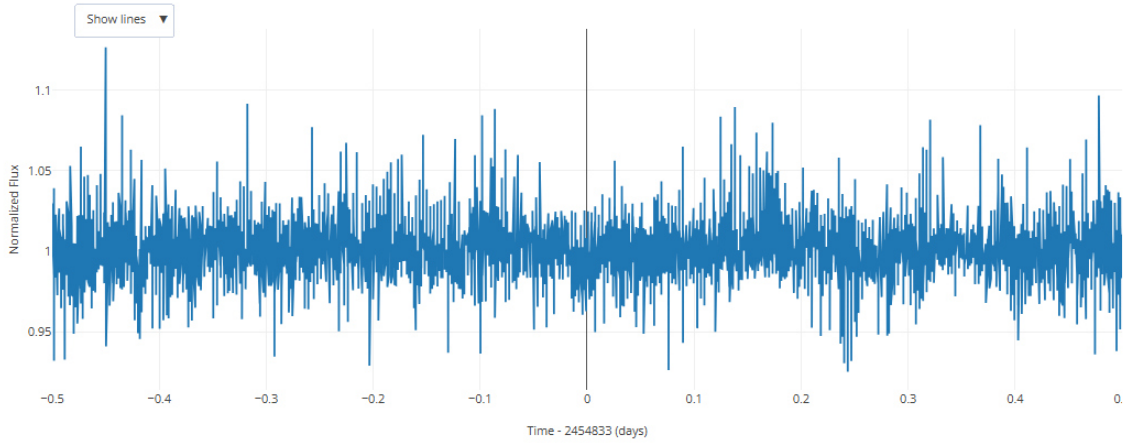
Plots



Output Values

Name	Value
Mission	K2

Input Values

Name	Value
Aperture Columns	5

Earth Comparison:

Property	Earth	TRAPPIST-1e
Radius (Earth Radii)	1	0.92
Mass (Earth Mass)	1	0.69
Density (g/cm3)	5.51	4.87
Orbital Period (days)	365	6.1
Temperature (K)	255	~250

NOTE: The data above is obtained/calculated from the base observed data from the NASA Archives, Source: "Agol et al. 2021" .

Analysis of TRAPPIST-1e based on the observed/calculated data

Density Analysis:

- Density of Earth = 5.51g/cm3
- TRAPPIST-1e density = 4.87 g/cm3

The normal range of density for Rocky or Terrestrial Planets is around 4-8 g/cm3 and the calculated density of TRAPPIST-1e is 4.87 g/cm3, therefore we may conclude that the planet is one which is Rocky/Terrestrial just like the Earth.

Conclusions:

From the above analysis we may conclude that TRAPPIST-1e is not a "Gas Giant", or not a planet like Neptune, but one which is similar to Earth.

Habitability:

TRAPPIST-1e orbits around a very cool dwarf star. The star is much smaller and cooler than Sun, thus its habitable zone is much closer, and the exoplanet is within the habitable zone.

- Earth Temperature = 255K
- TRAPPIST-e temperature = 250K

This suggests that temperatures at the planet is very much similar to that of Earth's.

This is a valuable comparison, which suggest that the planet could maintain water in liquid state.

Atmosphere:

As we have determined that the planet is NOT a gas giant, therefore we may rule out the hydrogen-rich atmosphere.

There is a potential for the planet to have a heavier atmosphere, one made of Nitrogen or CO2. With which we may consider the possibility of greenhouse effects which again suggests for the possibility of liquid-state water, oceans, and potential for life.

We could also expect a similar amount of stellar flux being received by the planet as that of Earth.

Bibliography

References:

1. Eric Agol *et al* 2021 *Planet. Sci. J.* **2** 1
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3. **NASA Exoplanet Archive. (2024).** *Planetary Systems Composite Parameters.* California Institute of Technology / IPAC. Available at: https://exoplanetarchive.ipac.caltech.edu/
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